

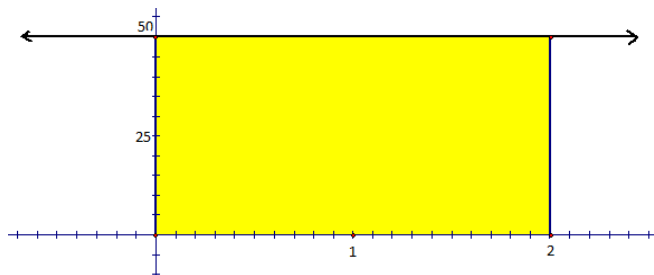
5.1 Approximating Areas

In Chapters 1 – 4 we have looked at the application of limits as it related to tangent lines and their slopes. In Chapter 5 turn our attention to a seeming unrelated question of using limits to help determine areas.

General Idea (Part I):

A car is traveling at a constant rate of 50mph for a 2-hour period. Find the total change in position. Represent this quantity as an area on a graph.

We can clearly see that $50 \text{ miles/hour} \times 2 \text{ hours} = 100 \text{ miles}$.



We can represent this graphically. We will have a horizontal axis representing *time* and a vertical axis representing *miles/hour*.

Notice that the area of the shaded rectangle is

$$2 \times 50 = 100$$

Note: When the rate of change of a quantity is positive, the total change in that quantity on some interval can be represented by the area between a function and the horizontal axis on that interval.

General Idea (Part II):

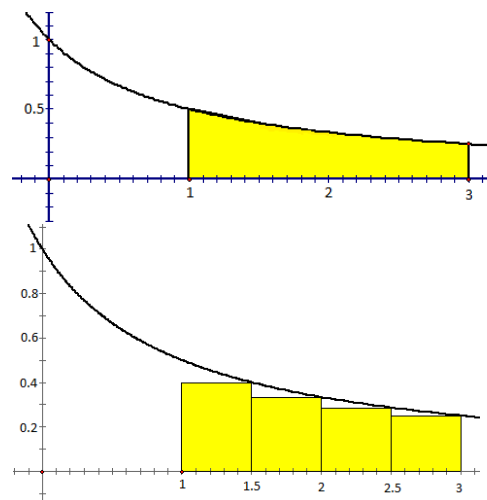
An object travels at $v(t) = \frac{1}{t+1} \text{ ft/sec}$ between the times $t = 1$ and $t = 3$ seconds. Estimate the total change in position of the object during this time.

Calculating this area will be much more difficult.

We can break up the interval $[1, 3]$ into smaller subintervals and we can then approximate the total change in position on each subinterval as the area of a rectangle.

The total change in position can be approximated by the sum of the areas of these rectangles.

$$\begin{aligned} A &\approx v(1.5) \cdot 0.5 + v(2) \cdot 0.5 + v(2.5) \cdot 0.5 + v(3) \cdot 0.5 \\ &= \frac{1}{2.5} \cdot 0.5 + \frac{1}{3} \cdot 0.5 + \frac{1}{3.5} \cdot 0.5 + \frac{1}{4} \cdot 0.5 \\ &= 0.5 \left[\frac{2}{5} + \frac{1}{3} + \frac{2}{7} + \frac{1}{4} \right] \\ &\approx 0.6345 \end{aligned}$$



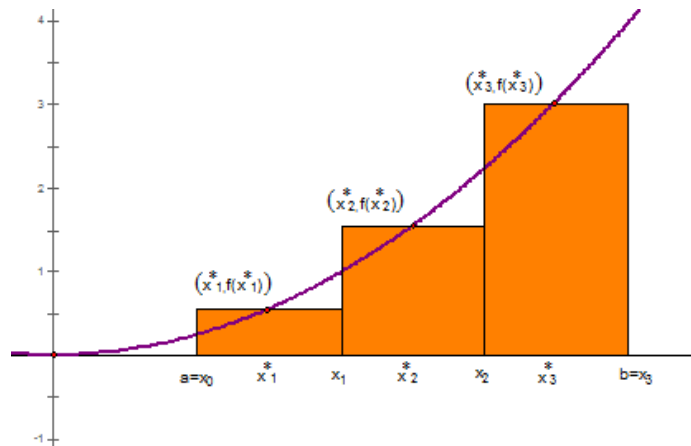
So we can say that the distance traveled by the object from $t = 1$ to $t = 3$ is approximately 0.6345 ft . This is an underestimate of the area (distance travelled).

Exploration: The area A of the region S that lies under the graph of the continuous function f can be approximated by using sum of areas of n approximating rectangles.

If

- each rectangle has an equal width $\Delta x = \frac{b-a}{n}$
- each rectangle has height $f(x_i^*)$, where x_i^* is any number in the i th subinterval $[x_{i-1}, x_i]$

Then the area of each rectangle will be given by the formula $f(x_i^*)\Delta x$.



Therefore the approximation of the area will be given by $A \approx \sum_{i=1}^n f(x_i^*)\Delta x$

Definition: The net area (also called signed area) of a region that lies between the x -axis and the graph of a continuous function f may be approximated with a Riemann sum which is a sum of the form

$$\sum_{i=1}^n f(x_i^*)\Delta x = f(x_1^*)\Delta x + f(x_2^*)\Delta x + f(x_3^*)\Delta x + \cdots + f(x_n^*)\Delta x \quad \text{where } x_i^* \text{ is a sample point in } [x_{i-1}, x_i].$$

Common Riemann Sums:

- A *Left Riemann Sum* has the top left corner of each rectangle on the function.
- A *Right Riemann Sum* has the top right corner of each rectangle on the function.
- A *Midpoint Riemann Sum* has the midpoint of the top side of each rectangle on the function.

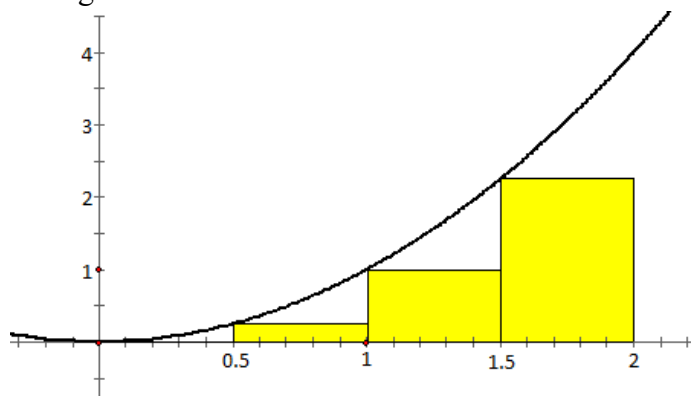
Example 1: Let A be the area of the region that lies above the x -axis and under the graph of $f(x) = x^2$ between $x = 0$ & $x = 2$.

a) Estimate the area by using a Left Riemann Sum with 4 rectangles.

Clearly the width of each rectangle is $\Delta x = \frac{2-0}{4} = \frac{1}{2}$.

Therefore,

$$\begin{aligned} A &\approx f(0) \cdot 0.5 + f(0.5) \cdot 0.5 + f(1) \cdot 0.5 + f(1.5) \cdot 0.5 \\ &= 0.5[f(0) + f(0.5) + f(1) + f(1.5)] \\ &= 0.5[0 + 0.25 + 1 + 2.25] \\ &= 1.75 \end{aligned}$$



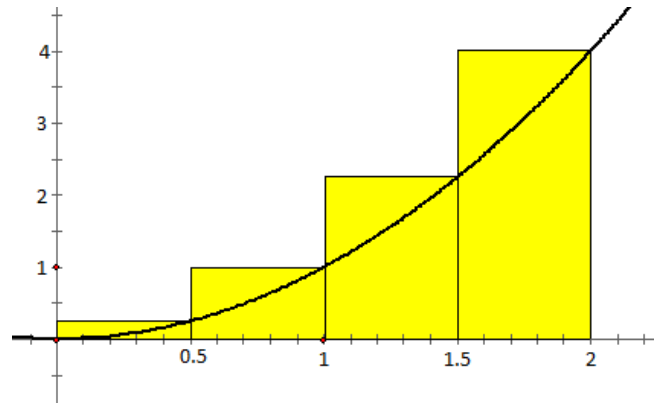
This is an underestimate of the area.
(Note: not all Left Riemann Sums are underestimates)

b) Estimate the area by using a Right Riemann Sum with 4 rectangles.

Clearly the width of each rectangle is $\Delta x = \frac{2-0}{4} = \frac{1}{2}$.

Therefore,

$$\begin{aligned} A &\approx f(0.5) \cdot 0.5 + f(1) \cdot 0.5 + f(1.5) \cdot 0.5 + f(2) \cdot 0.5 \\ &= 0.5[f(0.5) + f(1) + f(1.5) + f(2)] \\ &= 0.5[0.25 + 1 + 2.25 + 4] \\ &= 3.75 \end{aligned}$$



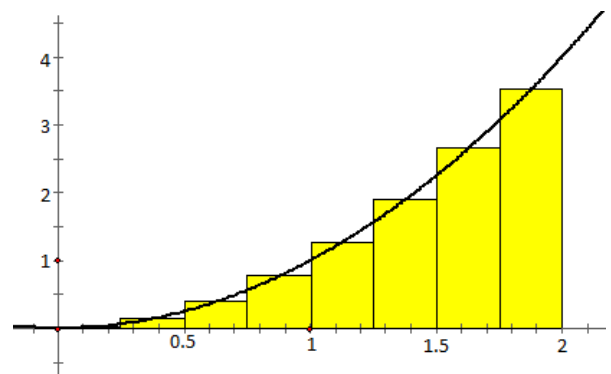
This is an overestimate of the area.
(Note: not all Right Riemann Sums are overestimates)

c) Estimate the area by using Midpoint Riemann Sum with 8 rectangles.

Clearly the width of each rectangle is $\Delta x = \frac{2-0}{8} = \frac{1}{4}$.

$$\text{Therefore, } A \approx \frac{1}{4} \left[f\left(\frac{1}{8}\right) + f\left(\frac{3}{8}\right) + f\left(\frac{5}{8}\right) + f\left(\frac{7}{8}\right) + f\left(\frac{9}{8}\right) + f\left(\frac{11}{8}\right) + f\left(\frac{13}{8}\right) + f\left(\frac{15}{8}\right) \right]$$

$$\begin{aligned} &= \frac{1}{4} \left[\frac{1}{64} + \frac{9}{64} + \frac{25}{64} + \frac{49}{64} + \frac{121}{64} + \frac{169}{64} + \frac{225}{64} \right] \\ &= \frac{1}{4} \left[\frac{680}{64} \right] \\ &= \frac{85}{32} \end{aligned}$$



It is difficult to tell if this is an overestimate or underestimate, so we don't label this as either.

Note: If f takes on both positive and negative values, then the rectangles above the x -axis have positive area and the rectangles below the x -axis have a negative area.

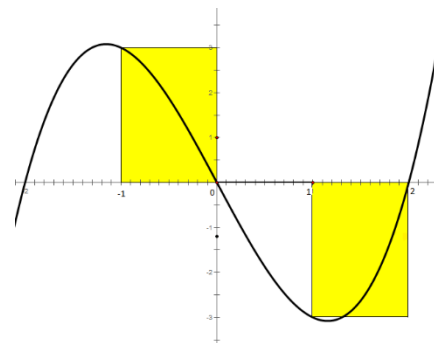
Example 2: Let A be the area of the region that lies between the x -axis and $f(x) = x^3 - 4x$ between $x = -1$ & $x = 2$.

a) Estimate the area by using a Riemann sum with left endpoints and 3 rectangles.

Clearly the width of each rectangle is $\Delta x = \frac{2 - (-1)}{3} = \frac{3}{3} = 1$.

Therefore,

$$\begin{aligned} A &\approx 1[f(-1) + f(0) + f(1)] \\ &= 1[3 + 0 - 3] \\ &= 1[0] \\ &= 0 \end{aligned}$$

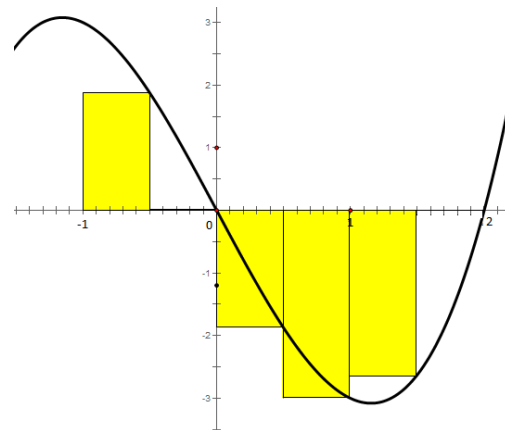


b) Estimate the area by using a Riemann sum with right endpoints and 6 rectangles.

Clearly the width of each rectangle is $\Delta x = \frac{2 - (-1)}{6} = \frac{3}{6} = \frac{1}{2}$.

Therefore,

$$\begin{aligned} A &\approx \frac{1}{2}[f(-0.5) + f(0) + f(0.5) + f(1) + f(1.5) + f(2)] \\ &= \frac{1}{2}\left[\frac{15}{8} + 0 - \frac{15}{8} - 3 - \frac{21}{8} - 0\right] \\ &= -\frac{45}{16} = -2.8125 \end{aligned}$$



Example 3: Suppose the odometer on our car is broken and we want to estimate the distance driven over a 30 minute time interval. We take the speedometer readings every 5 minutes $\left(\frac{1}{12} \text{ hr}\right)$ and record them in the following table:

Time (hours)	0	1/12	2/12	3/12	4/12	5/12	6/12
Velocity(mi/h)	17	21	24	29	32	31	28

a) Estimate the distance traveled using left endpoints.

$$\text{distance} = \frac{1}{12}[17 + 21 + 24 + 29 + 32 + 31] = \frac{1}{12}[154] = \frac{77}{6} \text{ miles} \approx 12.83 \text{ miles}$$

b) Estimate the distance traveled using right endpoints.

$$\text{distance} = \frac{1}{12}[21 + 24 + 29 + 32 + 31 + 28] = \frac{1}{12}[165] = 13.75 \text{ miles}$$